

Week 2

- **Problem 1**

Draw a Venn diagram for three sets X, Y, Z to represent each expression:

- (a) $(X \setminus Y) \cup Z$
- (b) $(X \cap Y)^c \setminus Z$

- **Solution 1.1**

Skip.

- **Problem 2**

Let $M = \{4, \{5\}, \{4, 5\}\}$. Determine which of the following assertions are correct.

- (a) $5 \in M$
- (c) $\{5\} \subseteq M$
- (b) $(A \setminus B)^c = A^c \cup B$
- (f) $A \subseteq B \iff A \cap B = A$
- (g) $A \cup (A \cap B) = A$
- (j) $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$
- (e) $\{4, 5\} \in M$
- (g) $\{\{4, 5\}\} \in M$
- (i) $\emptyset \in M$
- (k) $M \in M$

- **Solution 2.1**

- **(a) Incorrect.** The number 5 is not a direct element of M ; the set $\{5\}$ is.
- **(c) Incorrect.** For $\{5\} \subseteq M$ to be true, $5 \in M$ must hold, which is false.
- **(b) Correct.** By definition,

$$\begin{aligned}x \in (A \setminus B)^c &\iff x \notin A \setminus B \\&\iff \neg(x \in A \wedge x \notin B) \\&\iff x \notin A \vee x \in B \\&\iff x \in A^c \vee x \in B \\&\iff x \in A^c \cup B\end{aligned}$$

- **(f) Correct.**

($\implies$) If $A \subseteq B$, then for any $x \in A$, $x \in B$ must hold. Thus $x \in A \implies x \in A \wedge x \in B \implies x \in A \cap B$, showing $A \subseteq A \cap B$. Since $A \cap B \subseteq A$ always holds by definition, $A \cap B = A$.

($\impliedby$) If $A \cap B = A$, then for any $x \in A$, we have $x \in A \cap B$, which means $x \in B$ by definition of intersection. Thus $A \subseteq B$.

- **(g) Correct.** By definition, $x \in A \cup (A \cap B) \iff x \in A \vee (x \in A \wedge x \in B) \iff x \in A$.
- **(j) Correct.** By definition:

$$x \in A \cap (B \setminus D) \iff x \in A \wedge (x \in B \wedge x \notin D) \iff (x \in A \wedge x \in B) \wedge x \notin D$$

On the other side:

$$x \in (A \cap B) \setminus (A \cap D) \iff (x \in A \wedge x \in B) \wedge \neg(x \in A \wedge x \in D) \iff (x \in A \wedge x \in B) \wedge (x \notin A \vee x \notin D)$$

Since $x \in A$ is already true from the first part, $x \notin A$ is false, so $(x \notin A \vee x \notin D)$ simplifies directly to $x \notin D$. Thus, both sides mean $(x \in A \wedge x \in B) \wedge x \notin D$.

- **(e) Correct.** The set $\{4, 5\}$ is explicitly listed as the third element of M .
 - **(g) Incorrect.** The nested set $\{\{4, 5\}\}$ is not an element belonging to M .
 - **(i) Incorrect.** The empty set $\emptyset$ is not explicitly listed inside M .
 - **(k) Incorrect.** By the Axiom of Regularity, no set can contain itself as an element.
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- **Problem 3**

Suppose $A, B, D \subseteq U$. Prove the following properties.

- (a) $A \cap (A \cup B) = A$
- (c) $A \setminus (B \cup D) = (A \setminus B) \cap (A \setminus D)$
- (d) $(A \cap B)^c = A^c \cup B^c$
- (e) $A \subseteq B \iff A \cup B = B$
- (h) $(A \cup B)^c = A^c \cap B^c$
- (i) $(A \setminus B) \cup (A \cap B) = A$

◦ **Solution 3.1**

■ **(a) Proof:**

We prove this by showing $x \in A \cap (A \cup B) \iff x \in A$:

$$x \in A \cap (A \cup B) \iff x \in A \wedge (x \in A \vee x \in B) \iff x \in A$$

■ **(c) Proof:**

We use the element-hood definition to show logical equivalence:

$$\begin{aligned} x \in A \setminus (B \cup D) &\iff x \in A \wedge x \notin (B \cup D) \\ &\iff x \in A \wedge \neg(x \in B \vee x \in D) \\ &\iff x \in A \wedge (x \notin B \wedge x \notin D) \\ &\iff (x \in A \wedge x \notin B) \wedge (x \in A \wedge x \notin D) \\ &\iff x \in (A \setminus B) \wedge x \in (A \setminus D) \\ &\iff x \in (A \setminus B) \cap (A \setminus D) \end{aligned}$$

■ **(d) Proof:**

By definition of the complement and logical negation:

$$\begin{aligned} x \in (A \cap B)^c &\iff x \notin A \cap B \\ &\iff \neg(x \in A \wedge x \in B) \\ &\iff x \notin A \vee x \notin B \\ &\iff x \in A^c \vee x \in B^c \\ &\iff x \in A^c \cup B^c \end{aligned}$$

■ **(e) Proof:**

($\implies$) Assume $A \subseteq B$, meaning if $x \in A$ then $x \in B$. We show $A \cup B = B$:

- If $x \in A \cup B$, then $x \in A$ or $x \in B$. If $x \in A$, then $x \in B$ by assumption. Thus $x \in B$ in both cases, which means $A \cup B \subseteq B$.
- If $x \in B$, then $x \in A \vee x \in B$ is automatically true, so $x \in A \cup B$, meaning $B \subseteq A \cup B$. Thus, $A \cup B = B$.

($\impliedby$) Assume $A \cup B = B$. Let $x \in A$. Then $x \in A \vee x \in B$ is true, so $x \in A \cup B$. Since $A \cup B = B$, it follows that $x \in B$. Hence, $A \subseteq B$.

■ **(h) Proof:**

By definition of the complement and logical negation:

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \cup B \\ &\iff \neg(x \in A \vee x \in B) \\ &\iff x \notin A \wedge x \notin B \\ &\iff x \in A^c \wedge x \in B^c \\ &\iff x \in A^c \cap B^c \end{aligned}$$

■ **(i) Proof:**

We use the element-hood definition to show logical equivalence:

$$\begin{aligned} x \in (A \setminus B) \cup (A \cap B) &\iff x \in (A \setminus B) \vee x \in (A \cap B) \\ &\iff (x \in A \wedge x \notin B) \vee (x \in A \wedge x \in B) \\ &\iff x \in A \wedge (x \notin B \vee x \in B) \\ &\iff x \in A \wedge \text{True} \\ &\iff x \in A \end{aligned}$$

• **Problem 4**

Let A, B, C be subsets of a universal set X . Prove the implication: If $A \cap B \subseteq C$ and $C \subseteq A \cup B$, then $C \setminus A \subseteq B$.

◦ **Solution 4.1**

Proof:

Let $x \in C \setminus A$. By definition of set difference, this means:

$$x \in C \quad \text{and} \quad x \notin A$$

We are given the condition $C \subseteq A \cup B$. Since $x \in C$, it must follow by definition of subset that:

$$x \in A \cup B$$

By definition of union, this means $x \in A$ or $x \in B$.

However, we already established from the definition of $C \setminus A$ that $x \notin A$.

Therefore, for the disjunction ($x \in A \vee x \in B$) to hold true while $x \in A$ is false, $x \in B$ must be true.

Since any arbitrary element $x \in C \setminus A$ satisfies $x \in B$, we conclude by definition of subset that:

$$C \setminus A \subseteq B$$

The proof is complete.

• **Problem 5**

Let A, B be arbitrary sets. Prove or disprove: $\mathcal{P}(A \setminus B) = \mathcal{P}(A) \setminus \mathcal{P}(B)$. If false, provide an explicit counterexample.

◦ **Solution 5.1**

The statement is **false**.

Counterexample:

Let $A = \{1\}$ and $B = \{2\}$.

Then, the set difference is $A \setminus B = \{1\}$.

The power sets of these collections are:

$$\mathcal{P}(A \setminus B) = \{\emptyset, \{1\}\}$$

$$\mathcal{P}(A) = \{\emptyset, \{1\}\}, \quad \mathcal{P}(B) = \{\emptyset, \{2\}\}$$

Evaluating the right side of the proposed equation yields:

$$\mathcal{P}(A) \setminus \mathcal{P}(B) = \{\{1\}\}$$

Since $\{\emptyset, \{1\}\} \neq \{\{1\}\}$, the statement does not hold in general.

• **Problem 6**

Let $A, B \subseteq X$. Prove that $A \subseteq B$ if and only if for every set C , $A \cap C \subseteq B \cap C$.

◦ **Solution 6.1**

Proof:

($\Rightarrow$) Assume $A \subseteq B$. Let C be any arbitrary set, and let $x \in A \cap C$. By definition of intersection, this means:

$$x \in A \quad \text{and} \quad x \in C$$

Since $A \subseteq B$, by definition of subset, $x \in A \Rightarrow x \in B$. Therefore, we have:

$$x \in B \quad \text{and} \quad x \in C$$

By definition of intersection, $x \in B \cap C$. Since any arbitrary element $x \in A \cap C$ satisfies $x \in B \cap C$, we conclude that $A \cap C \subseteq B \cap C$.

($\Leftarrow$) Assume that for every set C , $A \cap C \subseteq B \cap C$ holds. We need to show $A \subseteq B$.

Let x be an arbitrary element such that $x \in A$.

Since the given hypothesis holds true for any choice of set C , we can choose C to be the singleton set containing only x , that is, $C = \{x\}$.

Now evaluate the intersection $A \cap \{x\}$: since $x \in A$ and $x \in \{x\}$, we have:

$$x \in A \cap \{x\}$$

By our hypothesis, $A \cap \{x\} \subseteq B \cap \{x\}$, which implies:

$$x \in B \cap \{x\}$$

By definition of intersection, this means $x \in B$ and $x \in \{x\}$. In particular, we obtain:

$$x \in B$$

Since $x \in A \Rightarrow x \in B$ for any arbitrary element, we conclude by definition of subset that $A \subseteq B$.

The proof is complete.

• **Problem 7**

Let U be a universal set, and let $A, B \subseteq U$. Prove or disprove: $(A \cup B)^c = A^c \cup B^c$. If the statement is false, give a counterexample.

◦ **Solution 7.1**

The statement is **false**.

Counterexample:

Let the universal set be $U = \{1, 2\}$, and let $A = \{1\}$ and $B = \{2\}$.

First, find the union of A and B :

$$A \cup B = \{1, 2\}$$

By definition of complement, the left-hand side is:

$$(A \cup B)^c = U \setminus \{1, 2\} = \emptyset$$

Now, calculate the complements of A and B separately:

$$A^c = \{2\}, \quad B^c = \{1\}$$

Thus, the right-hand side is:

$$A^c \cup B^c = \{1, 2\}$$

Since $\emptyset \neq \{1, 2\}$, the statement does not hold.

• **Problem 8**

Suppose $A \subseteq B \cup C$. Does it follow that $A \subseteq B$ or $A \subseteq C$? If not, construct a concrete counterexample.

◦ **Solution 8.1**

No, it **does not follow**.

Counterexample:

Let $A = \{1, 2\}$, $B = \{1\}$, and $C = \{2\}$.

First, evaluate the union of B and C :

$$B \cup C = \{1, 2\}$$

Since every element of A is in $B \cup C$, the condition $A \subseteq B \cup C$ is satisfied.

However, $A \not\subseteq B$ because $2 \in A$ but $2 \notin B$.

Similarly, $A \not\subseteq C$ because $1 \in A$ but $1 \notin C$.

Therefore, the implication is false.

• **Problem 9**

Let $A, B, C \subseteq X$. Prove: $(A \setminus B) \setminus C = A \setminus (B \cup C)$.

◦ **Solution 9.1**

Proof:

We show that an element x belongs to the left-hand side if and only if it belongs to the right-hand side using basic definition of set-membership:

$$\begin{aligned} x \in (A \setminus B) \setminus C &\iff x \in (A \setminus B) \wedge x \notin C \\ &\iff (x \in A \wedge x \notin B) \wedge x \notin C \\ &\iff x \in A \wedge (x \notin B \wedge x \notin C) \\ &\iff x \in A \wedge \neg(x \in B \vee x \in C) \\ &\iff x \in A \wedge x \notin (B \cup C) \\ &\iff x \in A \setminus (B \cup C) \end{aligned}$$

Since $x \in (A \setminus B) \setminus C \iff x \in A \setminus (B \cup C)$ is logically equivalent, the two sets are equal.

• **Problem 10**

Let $A, B, C \subseteq X$. Show that $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$, where Δ denotes symmetric difference: $X \Delta Y = (X \setminus Y) \cup (Y \setminus X)$.

◦ **Solution 10.1**

Proof:

By definition of symmetric difference, $B \Delta C = (B \setminus C) \cup (C \setminus B)$. We unpack the condition $x \in A \cap (B \Delta C)$ step by step:

$$\begin{aligned} x \in A \cap (B \Delta C) &\iff x \in A \wedge x \in (B \Delta C) \\ &\iff x \in A \wedge (x \in B \setminus C \vee x \in C \setminus B) \\ &\iff x \in A \wedge ((x \in B \wedge x \notin C) \vee (x \in C \wedge x \notin B)) \\ &\iff (x \in A \wedge x \in B \wedge x \notin C) \vee (x \in A \wedge x \in C \wedge x \notin B) \end{aligned}$$

Now, we unpack the right-hand side expression, $x \in (A \cap B) \Delta (A \cap C)$:

$$\begin{aligned} x \in (A \cap B) \Delta (A \cap C) &\iff x \in ((A \cap B) \setminus (A \cap C)) \vee x \in ((A \cap C) \setminus (A \cap B)) \\ &\iff (x \in A \cap B \wedge x \notin A \cap C) \vee (x \in A \cap C \wedge x \notin A \cap B) \\ &\iff ((x \in A \wedge x \in B) \wedge \neg(x \in A \wedge x \in C)) \vee ((x \in A \wedge x \in C) \wedge \neg(x \in A \wedge x \in B)) \\ &\iff ((x \in A \wedge x \in B) \wedge (x \notin A \vee x \notin C)) \vee ((x \in A \wedge x \in C) \wedge (x \notin A \vee x \notin B)) \end{aligned}$$

In the first component, since $x \in A$ is true, the statement $(x \notin A \vee x \notin C)$ simplifies directly to $x \notin C$.
 In the second component, since $x \in A$ is true, the statement $(x \notin A \vee x \notin B)$ simplifies directly to $x \notin B$.
 Substituting these simplifications back, we get:

$$(x \in A \wedge x \in B \wedge x \notin C) \vee (x \in A \wedge x \in C \wedge x \notin B)$$

Since both sides reduce to the exact same logical statement, $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$ is proven.

• **Problem 11**

Another Morgan's Law (Indexed Families): Let $\{A_i\}_{i \in I}$ be a family of sets. Prove:

$$\left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c$$

◦ **Solution 11.1**

Proof:

We apply the logical definitions of indexed intersection, union, and complement to show element-hood equivalence:

$$\begin{aligned} x \in \left(\bigcap_{i \in I} A_i \right)^c &\iff x \notin \bigcap_{i \in I} A_i \\ &\iff \neg (\forall i \in I, x \in A_i) \\ &\iff \exists i \in I, \neg(x \in A_i) \\ &\iff \exists i \in I, x \notin A_i \\ &\iff \exists i \in I, x \in A_i^c \\ &\iff x \in \bigcup_{i \in I} A_i^c \end{aligned}$$

Since the logical statements match at every point, the set identity holds true.

• **Problem 12**

Properties of a Relation

Let

$$A = \{1, 2, 3, 4\}$$

and

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 1)\}$$

Determine whether R is:

- (a) reflexive;
- (b) symmetric;
- (c) antisymmetric;
- (d) transitive.

◦ **Solution 12.1**

- **(a) Reflexive:** Yes. By definition, a relation R on A is reflexive if for every $x \in A$, $(x, x) \in R$. For $A = \{1, 2, 3, 4\}$, the pairs $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$ are all explicitly present in R .
- **(b) Symmetric:** Yes. By definition, a relation is symmetric if for any $x, y \in A$, $(x, y) \in R \implies (y, x) \in R$. The only distinct pairs in R are $(1, 2)$ and $(2, 1)$. Since $(1, 2) \in R \implies (2, 1) \in R$ and $(2, 1) \in R \implies (1, 2) \in R$, the condition holds.
- **(c) Antisymmetric:** No. By definition, a relation is antisymmetric if for any $x, y \in A$, $((x, y) \in R \wedge (y, x) \in R) \implies x = y$. In this relation, we have $(1, 2) \in R$ and $(2, 1) \in R$, but $1 \neq 2$.
- **(d) Transitive:** Yes. By definition, a relation is transitive if for any $x, y, z \in A$, $((x, y) \in R \wedge (y, z) \in R) \implies (x, z) \in R$. Checking the non-trivial combinations:

$$((1, 2) \in R \wedge (2, 1) \in R) \implies (1, 1) \in R$$

$$((2, 1) \in R \wedge (1, 2) \in R) \implies (2, 2) \in R$$

Any other compositions with diagonal relations automatically satisfy the implication.

• **Problem 13**

Relations on Integers

For each of the following relations on $\mathbb{Z}$, determine whether it is reflexive, symmetric, antisymmetric, and transitive. For every

property that fails, give a counterexample.

(a)

$$a \sim b \iff a = b$$

(b)

$$a \sim b \iff a \leq b$$

(c)

$$a \sim b \iff |a - b| \leq 2$$

◦ **Solution 13.1**

Part (a):

- **Reflexive:** Yes. For any integer $a \in \mathbb{Z}$, the statement $a = a$ is universally true. Therefore, $a \sim a$ holds for all $a \in \mathbb{Z}$.
- **Symmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$. By definition, this means $a = b$. Since equality is symmetric, it follows that $b = a$, which means $b \sim a$.
- **Antisymmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$ and $b \sim a$. By definition, this implies $a = b$ and $b = a$. This directly satisfies the requirement that $a = b$.
- **Transitive:** Yes. Let $a, b, c \in \mathbb{Z}$ such that $a \sim b$ and $b \sim c$. By definition, this yields $a = b$ and $b = c$. Substituting b gives $a = c$, which means $a \sim c$.

◦ **Solution 13.2**

Part (b):

- **Reflexive:** Yes. For any integer $a \in \mathbb{Z}$, the statement $a \leq a$ is always true. Thus, $a \sim a$ holds for all $a \in \mathbb{Z}$.
- **Symmetric:** No. Let $a = 1$ and $b = 2$. We have $1 \leq 2$, so $1 \sim 2$ is true. However, $2 \leq 1$ is false, so $2 \sim 1$ does not hold.
- **Antisymmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$ and $b \sim a$. By definition, this means $a \leq b$ and $b \leq a$. By the properties of the standard order relation on integers, $a \leq b \wedge b \leq a \implies a = b$.
- **Transitive:** Yes. Let $a, b, c \in \mathbb{Z}$ such that $a \sim b$ and $b \sim c$. By definition, this yields $a \leq b$ and $b \leq c$. By the transitivity property of inequalities, $a \leq b \wedge b \leq c \implies a \leq c$, which means $a \sim c$.

◦ **Solution 13.3**

Part (c):

- **Reflexive:** Yes. For any integer $a \in \mathbb{Z}$, we have $|a - a| = |0| = 0$. Since $0 \leq 2$, the condition $|a - a| \leq 2$ is satisfied, so $a \sim a$ holds for all $a \in \mathbb{Z}$.
- **Symmetric:** Yes. Let $a, b \in \mathbb{Z}$ such that $a \sim b$. By definition, $|a - b| \leq 2$. Since $|a - b| = |-(b - a)| = |b - a|$, we have $|b - a| \leq 2$, which means $b \sim a$.
- **Antisymmetric:** No. Let $a = 1$ and $b = 2$. We have $|1 - 2| = |-1| = 1 \leq 2$, so $1 \sim 2$ holds. Similarly, $|2 - 1| = |1| = 1 \leq 2$, so $2 \sim 1$ holds. However, $1 \neq 2$.
- **Transitive:** No. Let $a = 1$, $b = 3$, and $c = 5$. We have $|1 - 3| = |-2| = 2 \leq 2$, so $1 \sim 3$ holds. We also have $|3 - 5| = |-2| = 2 \leq 2$, so $3 \sim 5$ holds. However, for a and c , we have $|1 - 5| = |-4| = 4 > 2$, which means $1 \sim 5$ does not hold.

• **Problem 14**

An Equivalence Relation on Integers

Define a relation $\sim$ on $\mathbb{Z}$ by

$$a \sim b \iff a - b \text{ is divisible by } 4.$$

(a) Prove that $\sim$ is an equivalence relation.

(b) Find $[0]$, $[1]$, $[2]$, $[3]$, $[5]$.

(c) How many distinct equivalence classes are there?

◦ **Solution 14.1**

Part (a):

- **Reflexive:** Let $a \in \mathbb{Z}$. Then $a - a = 0$. Since $0 = 4 \times 0$, 0 is divisible by 4. Thus, $a \sim a$ for all $a \in \mathbb{Z}$.
- **Symmetric:** Let $a, b \in \mathbb{Z}$ such that $a \sim b$. By definition, $a - b = 4k$ for some $k \in \mathbb{Z}$. Then $b - a = -(a - b) = -4k = 4(-k)$. Since $-k \in \mathbb{Z}$, $b - a$ is divisible by 4, so $b \sim a$.
- **Transitive:** Let $a, b, c \in \mathbb{Z}$ such that $a \sim b$ and $b \sim c$. By definition, $a - b = 4k$ and $b - c = 4m$ for some $k, m \in \mathbb{Z}$. Adding these equations gives $(a - b) + (b - c) = 4k + 4m \implies a - c = 4(k + m)$. Since $k + m \in \mathbb{Z}$, $a - c$ is divisible by 4, so $a \sim c$.

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation.

◦ **Solution 14.2**

Part (b):

By definition, $[a] = \{x \in \mathbb{Z} \mid x - a = 4k, k \in \mathbb{Z}\} = \{a + 4k \mid k \in \mathbb{Z}\}$.

- $[0] = \{\dots, -8, -4, 0, 4, 8, \dots\} = \{4k \mid k \in \mathbb{Z}\}$
- $[1] = \{\dots, -7, -3, 1, 5, 9, \dots\} = \{1 + 4k \mid k \in \mathbb{Z}\}$
- $[2] = \{\dots, -6, -2, 2, 6, 10, \dots\} = \{2 + 4k \mid k \in \mathbb{Z}\}$
- $[3] = \{\dots, -5, -1, 3, 7, 11, \dots\} = \{3 + 4k \mid k \in \mathbb{Z}\}$
- $[5] = \{\dots, -3, 1, 5, 9, 13, \dots\} = [1]$

◦ **Solution 14.3**

Part (c):

There are exactly 4 distinct equivalence classes, which are $[0]$, $[1]$, $[2]$, and $[3]$.

• **Problem 15**

Same Remainder

Let

$$A = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

Define a relation $\sim$ on A by

$$a \sim b \iff a \text{ and } b \text{ have the same remainder when divided by 3.}$$

(a) Write down all equivalence classes.

(b) Write the relation $\sim$ explicitly as a set of ordered pairs.

◦ **Solution 15.1**

Part (a):

We group the elements of A by their remainder when divided by 3:

- Remainder 0: $[0] = \{0, 3, 6\}$
- Remainder 1: $[1] = \{1, 4, 7\}$
- Remainder 2: $[2] = \{2, 5\}$

◦ **Solution 15.2**

Part (b):

The relation $\sim$ consists of all pairs (x, y) such that x and y belong to the same equivalence class:

$$\begin{aligned} \sim = \{ & (0, 0), (0, 3), (0, 6), (3, 0), (3, 3), (3, 6), (6, 0), (6, 3), (6, 6), \\ & (1, 1), (1, 4), (1, 7), (4, 1), (4, 4), (4, 7), (7, 1), (7, 4), (7, 7), \\ & (2, 2), (2, 5), (5, 2), (5, 5) \} \end{aligned}$$

• **Problem 16**

A Partial Order on Sets

Let

$$X = \{1, 2, 3\}.$$

Consider the relation $\subseteq$ on $\mathcal{P}(X)$. Prove that $\subseteq$ is a partial order.

◦ **Solution 16.1**

Proof:

- **Reflexive:** Let $A \in \mathcal{P}(X)$. For any $x \in A$, the statement $x \in A$ is true, so $A \subseteq A$ holds by definition of subset.
- **Antisymmetric:** Let $A, B \in \mathcal{P}(X)$ such that $A \subseteq B$ and $B \subseteq A$. By definition of subset, $A \subseteq B \implies (\forall x \in A \implies x \in B)$ and $B \subseteq A \implies (\forall x \in B \implies x \in A)$. Thus $x \in A \iff x \in B$, which implies $A = B$.
- **Transitive:** Let $A, B, C \in \mathcal{P}(X)$ such that $A \subseteq B$ and $B \subseteq C$. Let $x \in A$. Since $A \subseteq B$, it follows that $x \in B$. Since $B \subseteq C$ and $x \in B$, it follows that $x \in C$. Thus $x \in A \implies x \in C$, meaning $A \subseteq C$. Since $\subseteq$ is reflexive, antisymmetric, and transitive, it is a partial order.

• **Problem 17**

Divisibility as a Partial Order

Let $\mathbb{N}$ denote the set of positive integers. Define a relation $\preceq$ on $\mathbb{N}$ by

$$x \preceq y \iff x \mid y.$$

Recall that $x \mid y$ means that there exists $k \in \mathbb{N}$ such that $y = xk$. Prove that $\preceq$ is a partial order on $\mathbb{N}$.

◦ **Solution 17.1**

Proof:

- **Reflexive:** Let $x \in \mathbb{N}$. We can write $x = x \times 1$. Since $1 \in \mathbb{N}$, there exists a positive integer $k = 1$ such that $x = xk$, meaning $x \mid x$. Thus, $x \preceq x$.
 - **Antisymmetric:** Let $x, y \in \mathbb{N}$ such that $x \preceq y$ and $y \preceq x$. By definition, $y = xk$ and $x = ym$ for some $k, m \in \mathbb{N}$. Substituting x into the first equation yields $y = (ym)k = y(mk)$. Dividing both sides by y (since $y \geq 1$) gives $1 = mk$. Since m, k are positive integers, the only solution is $m = 1$ and $k = 1$. Substituting $k = 1$ into $y = xk$ gives $y = x$.
 - **Transitive:** Let $x, y, z \in \mathbb{N}$ such that $x \preceq y$ and $y \preceq z$. By definition, $y = xk$ and $z = ym$ for some $k, m \in \mathbb{N}$. Substituting y into the equation for z gives $z = (xk)m = x(km)$. Since $k, m \in \mathbb{N}$, their product $km \in \mathbb{N}$. Thus, there exists an integer $k' = km$ such that $z = xk'$, meaning $x \mid z$, so $x \preceq z$.
- Since the relation $\preceq$ is reflexive, antisymmetric, and transitive, it is a partial order on $\mathbb{N}$.

• **Problem 18**

Inverse Relation

Let $\sim$ be a relation on a set X . The inverse relation $\sim^{-1}$ is defined by

$$x \sim^{-1} y \iff y \sim x.$$

Prove that $\sim$ is symmetric if and only if

$$\sim = \sim^{-1}.$$

◦ **Solution 18.1**

Proof:

($\implies$) Assume $\sim$ is symmetric. We prove $\sim = \sim^{-1}$ by showing mutual inclusion:

- Let $(x, y) \in \sim$, meaning $x \sim y$. Since $\sim$ is symmetric, we have $y \sim x$. By definition of the inverse relation, $y \sim x \iff x \sim^{-1} y$, which means $(x, y) \in \sim^{-1}$. Thus, $\sim \subseteq \sim^{-1}$.
 - Let $(x, y) \in \sim^{-1}$, meaning $x \sim^{-1} y$. By definition of the inverse relation, this means $y \sim x$. Since $\sim$ is symmetric, $y \sim x \implies x \sim y$, which means $(x, y) \in \sim$. Thus, $\sim^{-1} \subseteq \sim$.
- Therefore, $\sim = \sim^{-1}$.

($\impliedby$) Assume $\sim = \sim^{-1}$. To show $\sim$ is symmetric, let $x, y \in X$ such that $x \sim y$, which means $(x, y) \in \sim$.

Since $\sim = \sim^{-1}$, it follows that $(x, y) \in \sim^{-1}$, meaning $x \sim^{-1} y$.

By definition of the inverse relation, $x \sim^{-1} y \iff y \sim x$.

Thus, $x \sim y \implies y \sim x$, proving that $\sim$ is symmetric.

• **Problem 19**

Identity Relation and Antisymmetry

Let $\sim_X$ be the identity relation on X , defined by

$$x \sim_X y \iff x = y.$$

Let $\sim$ be a relation on X , and let $\sim^{-1}$ denote its inverse relation. Prove that $\sim$ is antisymmetric if and only if

$$\{(x, y) \in X \times X \mid x \sim y\} \cap \{(x, y) \in X \times X \mid x \sim^{-1} y\} \subseteq \{(x, y) \in X \times X \mid x \sim_X y\}.$$

◦ **Solution 19.1**

Proof:

For simplicity, let $R = \{(x, y) \in X \times X \mid x \sim y\}$, $R^{-1} = \{(x, y) \in X \times X \mid x \sim^{-1} y\}$, and $I = \{(x, y) \in X \times X \mid x \sim_X y\}$.

($\implies$) Assume $\sim$ is antisymmetric. Let $(x, y) \in R \cap R^{-1}$.

By definition of intersection, $(x, y) \in R$ and $(x, y) \in R^{-1}$.

This means $x \sim y$ and $x \sim^{-1} y$.

By definition of the inverse relation, $x \sim^{-1} y \iff y \sim x$.

Since $\sim$ is antisymmetric, $x \sim y \wedge y \sim x \implies x = y$.

By definition of the identity relation, $x = y \iff x \sim_X y \iff (x, y) \in I$.

Thus, $R \cap R^{-1} \subseteq I$.

($\impliedby$) Assume $R \cap R^{-1} \subseteq I$. To prove $\sim$ is antisymmetric, let $x, y \in X$ such that $x \sim y$ and $y \sim x$.

Since $x \sim y$, we have $(x, y) \in R$.

Since $y \sim x$, by definition of the inverse relation, we have $x \sim^{-1} y$, which means $(x, y) \in R^{-1}$.

Therefore, $(x, y) \in R \cap R^{-1}$.

By our inclusion hypothesis, $(x, y) \in R \cap R^{-1} \implies (x, y) \in I$.

By definition of I , $(x, y) \in I \iff x \sim_X y \iff x = y$.

Thus, $x \sim y \wedge y \sim x \implies x = y$, proving that $\sim$ is antisymmetric.

• **Problem 20**

Which Relations Define Functions?

For each of the following relations $R \subseteq \mathbb{R} \times \mathbb{R}$, determine whether R defines a function

$$f: \mathbb{R} \rightarrow \mathbb{R}.$$

Explain your answer.

(a) $R = \{(x, y) \in \mathbb{R}^2 : y = x^2\}.$

(b) $R = \{(x, y) \in \mathbb{R}^2 : y^2 = x\}.$

(c) $R = \{(x, y) \in \mathbb{R}^2 : xy = 1\}.$

(d) $R = \{(x, y) \in \mathbb{R}^2 : y^3 = x\}.$

◦ **Solution 20.1**

By definition, a relation $R \subseteq \mathbb{R} \times \mathbb{R}$ defines a function $f: \mathbb{R} \rightarrow \mathbb{R}$ if for every $x \in \mathbb{R}$, there exists a unique $y \in \mathbb{R}$ such that $(x, y) \in R$.

- **(a) Yes.** For every $x \in \mathbb{R}$, the value $y = x^2$ is a uniquely determined real number.
- **(b) No.** It fails on two conditions of the definition:
 - For $x = -1 \in \mathbb{R}$, there is no $y \in \mathbb{R}$ such that $y^2 = -1$.
 - For $x = 4 \in \mathbb{R}$, there is no unique y because both $y = 2$ and $y = -2$ satisfy $y^2 = 4$.
- **(c) No.** For $x = 0 \in \mathbb{R}$, there is no $y \in \mathbb{R}$ such that $0 \cdot y = 1$. Thus, $f(0)$ is undefined, so it cannot be a function from the entire domain $\mathbb{R}$.
- **(d) Yes.** For every $x \in \mathbb{R}$, every real number has a unique real cube root, so $y = \sqrt[3]{x}$ exists and is uniquely determined for all x .

• **Problem 21**

Natural Domains and Ranges

Verify the natural domains and corresponding ranges of the following functions. The domain in each case consists of all values of x for which the formula is well-defined.

Function	Domain	Range
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

◦ **Solution 21.1**

- **For $y = x^2$:**
 - **Domain:** The expression x^2 is defined for all real numbers, so the domain is $(-\infty, \infty)$.
 - **Range:** Since the square of any real number is non-negative, $x^2 \geq 0$, and for any $y \geq 0$, $x = \sqrt{y}$ exists. Thus, the range is $[0, \infty)$.
- **For $y = \frac{1}{x}$:**
 - **Domain:** The expression is well-defined as long as the denominator is non-zero, so $x \neq 0$, which gives $(-\infty, 0) \cup (0, \infty)$.
 - **Range:** Since $y = \frac{1}{x}$, y can never be 0. For any $y \neq 0$, we can choose $x = \frac{1}{y}$, so the range is $(-\infty, 0) \cup (0, \infty)$.
- **For $y = \sqrt{x}$:**
 - **Domain:** The square root is defined for non-negative real numbers, so $x \geq 0$, giving the domain $[0, \infty)$.
 - **Range:** The principal square root function always yields non-negative values, so $y \geq 0$, and for any $y \geq 0$, $x = y^2$ is in the domain. Thus, the range is $[0, \infty)$.
- **For $y = \sqrt{4 - x}$:**
 - **Domain:** We require $4 - x \geq 0 \iff x \leq 4$, which gives the interval $(-\infty, 4]$.

- **Range:** The square root outputs non-negative values, $y \geq 0$. For any $y \geq 0$, setting $y = \sqrt{4 - x}$ $\Leftrightarrow y^2 = 4 - x \Leftrightarrow x = 4 - y^2$, which is always ≤ 4 . Thus, the range is $[0, \infty)$.
- **For $y = \sqrt{1 - x^2}$:**
 - **Domain:** We require $1 - x^2 \geq 0 \Leftrightarrow x^2 \leq 1 \Leftrightarrow -1 \leq x \leq 1$, which gives the interval $[-1, 1]$.
 - **Range:** Since $-1 \leq x \leq 1$, we have $0 \leq x^2 \leq 1$, which implies $0 \leq 1 - x^2 \leq 1$. Taking the square root gives $0 \leq y \leq 1$. For any $y \in [0, 1]$, $x = \sqrt{1 - y^2}$ is a valid domain element. Thus, the range is $[0, 1]$.

• **Problem 22**

Domain, Codomain, and Image

Let

$$f: \mathbb{Z} \rightarrow \mathbb{Z}, \quad f(n) = 2n.$$

Determine:

- (a) the domain of f ;
- (b) the codomain of f ;
- (c) the image of f .

Is the image equal to the codomain?

◦ **Solution 22.1**

▪ **(a) Domain of f :**

By definition, for a function written as $f: A \rightarrow B$, the set A is the domain. Therefore, the domain of f is $\mathbb{Z}$ (the set of all integers).

▪ **(b) Codomain of f :**

By definition, for a function written as $f: A \rightarrow B$, the set B is the codomain. Therefore, the codomain of f is $\mathbb{Z}$ (the set of all integers).

▪ **(c) Image of f :**

The image of f is the set of all outputs generated by the function:

$$\text{Image}(f) = \{f(n) \mid n \in \mathbb{Z}\} = \{2n \mid n \in \mathbb{Z}\}$$

This is the set of all even integers, which can be written as $2\mathbb{Z} = \{\dots, -4, -2, 0, 2, 4, \dots\}$.

▪ **Is the image equal to the codomain?**

No. The image is the set of even integers $2\mathbb{Z}$, while the codomain is the set of all integers $\mathbb{Z}$. Since there are odd integers in the codomain (such as $1 \in \mathbb{Z}$) that do not belong to the image ($1 \notin 2\mathbb{Z}$), the image is a proper subset of the codomain, meaning they are not equal.

• **Problem 24**

Identity and Inclusion

Let

$$A = \{1, 2, 3\}, \quad B = \{1, 2, 3, 4\}.$$

Define

$$\text{id}_A: A \rightarrow A, \quad \text{id}_A(x) = x,$$

and the inclusion map

$$\iota: A \rightarrow B, \quad \iota(x) = x.$$

- (a) Find all values of id_A and ι .
- (b) Do the two functions have the same rule?
- (c) Are they equal as functions? Explain.

◦ **Solution 24.1**

- **(a)** By evaluating each element of the domain $A = \{1, 2, 3\}$ under their respective definitions:

$$\text{id}_A(1) = 1, \quad \text{id}_A(2) = 2, \quad \text{id}_A(3) = 3$$

$$\iota(1) = 1, \quad \iota(2) = 2, \quad \iota(3) = 3$$

- **(b) Yes.** Both functions have the exact same assignment rule, which maps any input element x directly to itself without alteration.

- **(c) No.** By definition, two functions are equal if and only if their domains, codomains, and rules are all identical. Here, the codomain of id_A is $A = \{1, 2, 3\}$, whereas the codomain of ι is $B = \{1, 2, 3, 4\}$. Since their codomains are not the same set ($A \neq B$), they are not equal as functions.

• **Problem 25**

Restriction of a Function

Let

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = x^2 - 1.$$

Let

$$A = [-1, 1]$$

and define

$$g = f|_A.$$

- (a) Write the domain and codomain of g .
- (b) Find $g(-1)$, $g(0)$, $g(1)$.
- (c) Are f and g equal?
- (d) Do they have the same formula?

◦ **Solution 25.1**

- **(a)** By definition of a function restriction, restricting a function $f: X \rightarrow Y$ to a subset $A \subseteq X$ means the new function has domain A while retaining the original rule and codomain. Therefore:

- Domain of g is $A = [-1, 1]$.

- Codomain of g is $\mathbb{R}$.

- **(b)** Since $g(x) = f(x)$ for all $x \in A$, we substitute the values into the formula $x^2 - 1$:

$$g(-1) = (-1)^2 - 1 = 1 - 1 = 0$$

$$g(0) = (0)^2 - 1 = 0 - 1 = -1$$

$$g(1) = (1)^2 - 1 = 1 - 1 = 0$$

- **(c) No.** By definition, two functions are equal if and only if they share the exact same domain, codomain, and assignment rule. Here, the domain of f is $\mathbb{R}$ while the domain of g is $[-1, 1]$. Since their domains are different ($\mathbb{R} \neq [-1, 1]$), the functions are not equal.
- **(d) Yes.** Both functions use the exact same algebraic formula $x^2 - 1$ to compute outputs from inputs, although g is only allowed to apply this formula to elements within the restricted set A .

• **Problem 26**

Construct Your Own Relation

Let

$$A = \{1, 2, 3, 4\}.$$

Construct:

- (a) a relation that is reflexive but not symmetric;
- (b) a relation that is symmetric but not reflexive;
- (c) an equivalence relation with exactly two equivalence classes;
- (d) a partial order different from the equality relation.

◦ **Solution 26.1**

- **(a)** To be reflexive, the relation must contain $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. To violate symmetry, we add a directional pair without its inverse, such as $(1, 2)$ without $(2, 1)$:

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2)\}$$

- **(b)** To violate reflexivity, at least one diagonal element must be missing (e.g., $(1, 1) \notin R$). To maintain symmetry, any ordered pair must have its coordinates flipped within the set:

$$R = \{(1, 2), (2, 1)\}$$

- **(c)** To form an equivalence relation with exactly two classes, we can partition A into two disjoint non-empty subsets, for instance, $A_1 = \{1, 2\}$ and $A_2 = \{3, 4\}$. The relation is constructed by taking $(A_1 \times A_1) \cup (A_2 \times A_2)$:

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}$$

Explicitly rewritten for verification:

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (3, 4), (4, 3), (4, 4)\}$$

- **(d)** A partial order must be reflexive, antisymmetric, and transitive. The standard total order relation "less than or equal to" on integers satisfies this. Restricting it to A yields:

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

This relation is distinctly different from the pure equality relation because it contains pairs where $x \neq y$ (such as $(1, 2)$).